

**THE NINTH WILLIAM LOWELL PUTNAM
MATHEMATICAL COMPETITION**

March 26, 1949

Morning Session

1. Answer either (i) or (ii):

(i) Three straight lines pass through the three points $(0, -a, a)$, $(a, 0, -a)$, and $(-a, a, 0)$, parallel to the x -axis, y -axis, and z -axis, respectively; $a > 0$. A variable straight line moves so that it has one point in common with each of the three given straight lines. Find the equation of the surface described by the variable line. (page 264)

(ii) Which planes cut the surface $xy + xz + yz = 0$ in (1) circles, (2) parabolas? (page 266)

2. We consider three vectors drawn from the same initial point O , of lengths a , b , and c , respectively. Let E be the parallelepiped with vertex O of which the given vectors are the edges and H the parallelepiped with vertex O of which these vectors are the altitudes. Show that the product of the volumes of E and H equals $(abc)^2$, and generalize the theorem, with proof, to n dimensions. (page 268)

3. Assume that the complex numbers $a_1, a_2, \dots, a_n, \dots$ are all different from zero, and that $|a_r - a_s| > 1$ for $r \neq s$. Show that the series

$$\sum_{n=1}^{\infty} \frac{1}{a_n^3}$$

converges.

(page 269)

4. Given that P is a point inside a tetrahedron with vertices at A, B, C , and D , such that the sum of the distances $PA + PB + PC + PD$ is a minimum, show that the two angles $\angle APB$ and $\angle CPD$ are equal and are bisected by the same straight line. What other pairs of angles must be equal? (page 270)

5. How many roots of the equation $z^6 + 6z + 10 = 0$ lie in each quadrant of the complex plane? (page 272)

6. Prove that for every real or complex x

$$\prod_{k=1}^{\infty} \frac{1 + 2 \cos \frac{2x}{3^k}}{3} = \frac{\sin x}{x}. \quad (\text{page 274})$$

Afternoon Session

1. Each rational number p/q (p, q relatively prime positive integers) of the open interval $(0, 1)$ is covered by a closed interval of length $1/2q^2$, whose center is at p/q . Prove that $\sqrt{2}/2$ is not covered by any of the above closed intervals. (page 275)

2. Answer either (i) or (ii):

(i) Prove that

$$\sum_{n=2}^{\infty} \frac{\cos (\log \log n)}{\log n}$$

diverges.

(page 277)

(ii) Assume that $p > 0$, $a > 0$, and $ac - b^2 > 0$, and show that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{dx \, dy}{(p + ax^2 + 2bxy + cy^2)^2} = \pi p^{-1} (ac - b^2)^{-1/2}.$$

(page 278)

3. Let K be a closed plane curve such that the distance between any two points of K is always less than 1. Show that K lies inside a circle of radius $1/\sqrt{3}$. (page 279)

4. Show that the coefficients $a_1, a_2, a_3, \dots$ in the expansion $\frac{1}{4} [1 + x - (1 - 6x + x^2)^{1/2}] = a_1x + a_2x^2 + a_3x^3 + \dots$ are positive integers. (page 282)

5. Let $a_1, a_2, \dots, a_n, \dots$ be an arbitrary sequence of positive numbers. Show that

$$\limsup_{n \rightarrow \infty} \left(\frac{a_1 + a_{n+1}}{a_n} \right)^n \geq e. \quad (\text{page 283})$$

6. Let C be a closed convex curve with a continuously turning tangent and let O be a point inside C . With each point P on C we associate the point $T(P)$ on C which is defined as follows: Draw the tangent to C at P and from O drop the perpendicular to that tangent. $T(P)$ is then the point at which this perpendicular intersects the curve C .

Starting now with a point P_0 on C , we define points P_n by the formula $P_n = T(P_{n-1})$, $n \geq 1$. Prove that the points P_n approach a limit, and characterize those points which can be limits of sequences P_n . (You may consider the facts that T is a continuous transformation and that a convex curve lies on one side of each of its tangents as not requiring proofs.) (page 285)